

Khair (Catechu) is a tree of sub-tropical forests. These types of forests are found in small shrubs. This forest is found in the Shivalik mountains and the western Himalayas at an altitude of more than 1000 meters. Major species found in these forests are – khair, keekar, Acacia etc.

38. Which one of the following States has the highest percentage of area under forests?

- (a) Himachal Pradesh (b) Assam
(c) Andhra Pradesh (d) Arunachal Pradesh
(e) None of the above / More than one of the above

63rd B.P.C.S. (Pre) 2017

Ans. (e)

According to the India State of Forest Report, 2023, the state of Mizoram has the highest percentage of area under forest.

39. which tree is found in plurality in Sita Nadi area?

- (a) Bamboo (b) Teak
(c) Mahua (d) Sal

Chhattisgarh P.C.S. (Pre), 2022

Ans. (d)

The topography of the Sita Nadi sanctuary is steep and highly variable, covering an area of 553.36 sq. km. The Sita Nadi Sanctuary is famous for its abundant flora and is home to Sal, Teak and Bamboo forests. These Sal forests in the Sita Nadi Sanctuary are considered as the best in the country.

Irrigation and Canals

*India is situated in tropical and sub-tropical climate area. The maximum area of the country comes under a high-temperature zone, except for the western Himalayas, north-eastern-Assam, Meghalaya and some parts of South India. *High evaporation occurs due to high temperatures. Therefore, there is a need for irrigation to fulfill the deficit of moisture in sown crops. *In 2021-22 (P), tubewells were the primary source of irrigation in the country. **47.3%** of total net irrigated area is irrigated by tubewells. Both tubewells and wells irrigate **60.5%** of the total net irrigated area of the country. ***24.7%** of net irrigated land is irrigated by canals and **2.8%** by tanks. The remaining irrigated land is irrigated by other means. Indian irrigation projects can be classified into three groups.

(i) **Minor irrigation projects - Up to 2000 hectares of** are irrigated under these projects. These include wells, tubewells, ponds, pumpsets, drip irrigation, sprinklers etc. Minor irrigation projects fulfill about **62%** of the demand

for irrigation in India.

(ii) **Medium irrigation projects - More than 2000 to 10000 hectares of** area are irrigated under these projects.

(iii) **Major irrigation projects - More than 10,000 hectares of area are irrigated under these projects. 38% of the need for irrigation in the country is fulfilled by the major irrigation projects.**

***Gang Canal** is one of the world's oldest and most developed canals. It was constructed in the year 1927 by **Shri Gang Singh**, the then Maharaja of Bikaner. It arose from Hussaini Wala, near the river Sutlej from Firojpur. *The **Eastern Yamuna Canal** originates from the left bank of Yamuna at Tajewala. The length of the main canal is **197** kilometres. This canal system provides irrigation facility to Hindon, Yamuna Doab's, Saharanpur, Muzaffarnagar, Meerut and Ghaziabad. After its revival, the canal started functioning in 1830 A.D. At that time, the water discharged from this canal was about 800 cusecs. In the meantime, this was increased to about 4000 cusecs. To solve this problem, the Hathinikund BNarrage (Haryana) has been constructed about 3 km from Tajewala. *The construction of **Upper Ganga Canal** was completed during the period of **1842 AD**. Sir P.T. Katley was the father of this project. The water was discharged for the first time on April 8, 1854, in this canal. It starts from a place called Bhimgoda in Haridwar on the right bank of the river Ganga. The length of the main canal is 298 km and the total length of this canal system is **6496 km**. Irrigation facilities are being provided in Haridwar, Saharanpur, Muzaffarnagar, Meerut, Ghaziabad, Bulandshahr, Aligarh, Etah, Mathura, Firozabad, Mainpuri, and Agra districts by this canal system. Initially, this canal system was also used for **waterways**, in addition to irrigation. 200 cusecs of water from this canal system are provided to Delhi and about 100 cusecs water is given to Harduwaganj Thermal Power Plant. *The **sharda Major Canal** has been constructed on the right bank of river Sarda near Banbasa in Nainital in order to provide protective irrigation facilities to the districts of Pilibhit, Bareilly, Lakhimpur Kheri, Shahjahanpur, Hardoi, Unnao, Lucknow, Barabanki, Raebareli, Pratapgarh, Sultanpur, Jaunpur, Azamgarh, Ghazipur and Prayagraj. The length of the main canal is **44.3** km and the total length of the Sharda canal system is **9961.3** km.

The **lower Ganga canal** starts from Narora (Bulandshahr).

Its construction work was completed in 1878 A.D. The length of the main canal is 98.8 km. and the total length of the canal system is 8278 km. *The foundation stone of the **Indira Gandhi canal** project was laid by the then Home Minister Govind Ballabh Pant in March, 1958. It originates from **Harike Dam** situated at the confluence of the Sutlej and Beas rivers in Punjab. It is one of the world's largest canal projects, with a total length of 649 km.

The canal mainly irrigates Sri Ganganagar, Bikaner, Jodhpur and Jaisalmer districts of western Rajasthan. *Under the **Gandak project** a 740 meter long barrage was built near the Triveni Ghat on Gandak River.

*The construction of **Farakka Barrage** was completed in the year 1975, with the aim of stopping the accumulation of silts in the Hooghly river. 40,000 cusecs of water from the Ganga river was diverted towards Hooghly river by a canal derived from the right bank of this barrage.

*Ten irrigation projects have been completed on the Bharatpuzha river and its tributaries in the Palakkad district of Kerala. The **Mangalam irrigation project** is one of them.

*The Garland **Canal system** was proposed by **Captain Dinshaw J. Dasthur**, The whereas the idea of linking rivers was first proposed by irrigation engineer **Vishweshwarya**. Later, the name of **K.L. Rao** was added to it. The then Prime Minister Atal Bihari Vajpayee inaugurated the development plan "**Hariyali**", related to water storage on January 27, 2003. Under this project, people are encouraged to conserve water in rural areas.

*On July 1, 2015, the **Pradhan Mantri Krishi Sichayi Yojana** (PMKSY) was approved to reduce the dependence of agriculture on monsoon and provide irrigation facilities to every farm. The Rural Development Ministry of the Government of India introduced three watershed development programmes untill April 1, 2008. These programmes are the **Integrated Wastelands Development Programme**, The **Drought Prone Areas Programme** and the **Desert Development Programme**. These programmes were brought under a comprehensive programme known as **Integrated Watershed Management Programme**.

*The main source of irrigation in Peninsular India is **Pond**. Since ancient times, ponds have been used for irrigation in these areas. The land of the Peninsular region is rocky; therefore, it is not possible to construct canals and wells easily. Besides, most of the rivers of Peninsular India are seasonal, which leads to a lack of water availability for canals. In this case, ponds serve as the main means of irrigation. Uttar

Pradesh (41.29 billion cubic m.), Punjab (32.80 billion cubic m.), Madhya Pradesh (17.33 billion cubic m.) and Maharashtra (15.29 billion cubic m.) are major states of India in terms of current annual groundwater extraction available for irrigation.

*According to the Statistical Diary Uttar Pradesh 2023 in the Uttar Pradesh (in 2021-22), in percentage of the net irrigated area by various means is **Tubewell** (73.9%), **Canal** (15.8%), **Well** (9.1%), **Tanks and Lakes** (0.6%) and by **other means** (0.6%). *Tubewell is the most widespread means of irrigation in Uttar Pradesh.

1. Pattiseema Project is associated with the integration of which of the following rivers?

- (a) Krishna and Kaveri
- (b) Krishna and Godavari
- (c) Godavari and Mahanadi
- (d) Ganga and Brahmaputra
- (e) None of the above / More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (b)

Pattiseema Lift Irrigation Project is a river interlinking project that connects Godavari River to Krishna River.

2. Which of the following is NOT correctly matched? (Canal Project) (Location)

- | | |
|------------------------------|-------------|
| (a) Ghatprabha Canal Project | Karnataka |
| (b) Ukai Canal Project | Gujarat |
| (c) Nira Canal Project | Maharashtra |
| (d) Jawai Project | Tamil Nadu |

U.P.B.E.O. (Pre) 2019

Ans. (d)

The Jawai Project is on the Jawai River, a tributary of Luni in Rajasthan. On the other hand, the Ukai Canal Project was undertaken by constructing a dam across the Tapti River in Gujarat. The Ghatprabha Canal Project is located in Karnataka on Ghatprabha river and the Nira Canal Project is in Maharashtra, on Nira river, a tributary of Bhima.

3. Given below are two statements. One is Assertion (A) and the other is Reason (R) :

Assertion (A) : One of the major sources of irrigation in Peninsular India is a pond.

Reason (A) : Most of the rivers in the Peninsular region are seasonal.

Code :

- (a) Both (A) and (R) are correct and (R) explains (A).
- (b) Both (A) and (R) are correct but (R) does not explain (A).

(c) (A) is correct, but (R) is false.

(d) (A) is false, but (R) is correct.

U.P. Lower Sub. (Pre) 2015

Ans. (a)

One of the major sources of irrigation in Peninsular India is pond irrigation as undulating relief and hard rock make it difficult to dig canals and wells. Thus assertion (A) is correct. Most of the rivers in Peninsular region are seasonal and dry up during summer season. So, they cannot supply water to canals. Therefore, the main source for irrigation is a pond. Thus, reason (R) also explains assertion (A).

4. With reference to India, which one of the following statements is not correct?

- (a) About one-third of the area of the country records more than 750 millimetres of annual rainfall
- (b) The dominant source of irrigation in the country are wells
- (c) Alluvial soil is the predominant type of soil in Northern plains of the country
- (d) The mountain areas account for about thirty percent of the surface area of the country.

I.A.S. (Pre) 2002

Ans. (b)

According to data for the year 2021-22, the major source of irrigation in the country is tubewells, which contribute 47.3% to the total net irrigated area. Tubewells and other wells provide for the irrigation of 60.5% of the net irrigated area. Canals irrigate 24.7% and tanks 2.8% of the total net irrigated area. The remaining land is dependent on other means of irrigation.

5. In which of the following states, the percentage of total cropped area under irrigation (2011) is the highest?

- (a) Madhya Pradesh
- (b) Maharashtra
- (c) Uttar Pradesh
- (d) Haryana

U.P.P.C.S. (Mains) 2015

Ans. (d)

The cropped area under irrigation (2011) is as follows –

States	2011	2015-16
Haryana	83.90%	96.91%
Uttar Pradesh	72.76%	83.17%
Madhya Pradesh	28.20%	56.25%
Maharashtra	16.78%	17.03%

Note - According to the Agriculture Census, 2015-16 data, the highest percentage of cropped area under irrigation in of Chandigarh (100%). Punjab is in second place (99.6%).

6. Which State in India has the largest extent in irrigation (percent)?

- (a) Punjab
- (b) Uttar Pradesh
- (c) Haryana
- (d) Madhya Pradesh

Uttarakhand P.C.S. (Pre) 2016

Ans. (a)

See the explanation of above the question.

7. With reference to micro-irrigation, which of the following statements is/are correct?

- 1. Fertilizer/nutrient loss can be reduced.
- 2. It is the only means of irrigation in dry land farming.
- 3. In some areas of farming, receding of groundwater level can be checked.

Select the correct answer using the code given below:

- (a) 1 only
- (b) 2 and 3
- (c) 1 and 3
- (d) 1, 2 and 3

I.A.S. (Pre) 2011

Ans. (c)

Fertilizer/nutrient loss can be reduced by micro-irrigation. In some areas of farming, receding of groundwater level can be checked by micro - irrigation due to limited and beneficial use of water. Thus, statements 1& 3 are correct while statement 2 is not correct.

8. Life saving or Protective irrigation refers to:

- (a) Constant flow of water
- (b) Irrigation at critical growth stage
- (c) P.W.P. Irrigation
- (d) Alternate wetting and drying

U.P.U.D.A./L.D.A. (Mains) 2010

Ans. (c)

Life-saving or Protective irrigation refers to irrigation at P.W.P (Permanent Wilting Point). The permanent wilting point is the extent at which the moisture is no longer available in sufficient quantity for plants to sustain themselves. Irrigation becomes compulsory in this situation.

9. During last 25 years, the development of tubewell irrigation has been most spectacular in :

- (a) Ganga-Ghaghra Doab
- (b) Ganga-Yamuna Doab
- (c) Saryupar Plain
- (d) Bundelkhand

U.P.P.C.S. (Pre) 1995

Ans. (c)

The development of tubewell irrigation has been most spectacular in the Saryupar Plain, due to lack of canals. Hence irrigation in this area is done by tubewells. Thus, the correct answer is option (c).

10. In which state of India, the maximum area is irrigated by tubewells?

- (a) Madhya Pradesh (b) Bihar
(c) Rajasthan (d) Uttar Pradesh

U.P.P.C.S. (Pre) 2006

Ans. (d)

In the states listed above, the most irrigated area by tubewells is in Uttar Pradesh. According to Statistical Diary of Uttar Pradesh 2023, Uttar Pradesh has 73.9% of its irrigated land area served by tubewells.

11. In which of the following States, the area under Tube-well irrigation is the largest?

- (a) Bihar (b) Haryana
(c) Uttar Pradesh (d) Punjab

U.P.P.C.S. (Pre) 2016

Ans. (c)

The net area under irrigation and the area irrigated by tubewells for the above states for the year 2018-19 (in thousand hectares) is as follows –

State	Net Irrigation Area (2018-19)	Net Area under tubewells irrigation (2018-19)
Uttar Pradesh	14392	10743
Punjab	4111	2944
Bihar	3115	1957
Haryana	3273	2066

12. The State having the largest area under tubewell and well irrigation in India is

- (a) Uttar Pradesh (b) Madhya Pradesh
(c) Andhra Pradesh (d) Punjab

U.P. P.C.S. (Mains) 2017

Ans. (a)

The state having the largest area under tubewell and well irrigation in India is Uttar Pradesh. According to the Statistical Diary of Uttar Pradesh 2023, in Uttar Pradesh, 10299 thousand hectares are irrigated by tubewells and 1269 thousand hectares are irrigated by wells.

13. Which one of the following is the correct sequence of the States (labelled 1, 2, 3 and 4) of India shown on the map in descending order in terms of their available groundwater resources for irrigation?



- (a) 3, 4, 1, 2 (b) 3, 4, 2, 1
(c) 4, 3, 1, 2 (d) 4, 3, 2, 1

I.A.S. (Pre) 1998

Ans. (d)

Annual groundwater extraction available for irrigation to the states in the given options is as follows –

Uttar Pradesh (4) – 41.29 Billion Cubic meters
Maharashtra (3) – 15.29 Billion Cubic meters
West Bengal (2) – 10.84 Billion Cubic meters
Assam (1) – 1.97 Billion Cubic meters

Thus, the descending order would be – Uttar Pradesh, Maharashtra, West Bengal and Assam.

14. The Garland Canal System in India was proposed by :

- (a) K.L. Rao (b) Dineshaw J. Dastur
(c) Visvesvaraya (d) Y.K. Alagh

U.P.P.C.S. (Pre) 1999

Ans. (b)

The Garland Canal System in India was proposed by Captain Dineshaw J. Dastur. The thought of river interlinking was first proposed by irrigation engineer M. Visvesvaraya. Later K.L. Rao also proposed another project of river interlinking. Then, the Water Resource Ministry rejected both projects (river interlinking & the Garland Canal System).

15. Who is regarded as the pioneer of Irrigation works in South India?

- (a) Sir Arthur Cotton (b) Col. Baird Smith
(c) Lt. Blaine (d) Col. Robert Smith

Uttarakhand P.C.S. (Pre) 2016

Ans. (a)

Sir Arthur Cotton was a British irrigation engineer who devoted his life to the construction of irrigation and navigation canal throughout British India. He is regarded as the pioneer of irrigation works in South India.

16. The maximum irrigation potential of India is created through:

- (a) Major Projects
- (b) Minor Projects & Major Projects
- (c) Minor Projects
- (d) Medium Projects

52nd B.P.S.C. (Pre) 2008

Ans. (c)

Irrigation projects in India are classified into three categories:

- 1. Minor Irrigation Projects:** All groundwater and surface water schemes that have a Culturable Command Area (CCA) up to 2,000 hectares individually are classified as Minor Irrigation Schemes. The minor irrigation projects comprise all groundwater development schemes such as dug wells, private shallow tubewells, deep public tubewells, and boring and deepening of dug wells, and small surface water development works such as storage tanks, lift irrigation projects, etc. 62 percent of the irrigation potential of India is created through minor irrigation projects.
- 2. Medium Irrigation Projects:** Those having a CCA of more than 2,000 hectares to 10,000 hectares fall under the category of medium irrigation projects.
- 3. Major Irrigation Projects:** Irrigation projects having a Culturable Command Area (CCA) of more than 10,000 hectares each are classified as major projects. 38 percent of the irrigation potential of India is created through major and medium irrigation projects.

17. The water-carrying capacity of Farakka canal is :

- (a) 75,000 Cusecs
- (b) 70,000 Cusecs
- (c) 40,000 Cusecs
- (d) 35,000 Cusecs

I.A.S. (Pre) 1997

Ans. (c)

The Farakka Barrage Project with headquarters at Farakka in Murshidabad, West Bengal was designed to serve the need for preservation and maintenance of Kolkata Port (Now-Syama Prasad Mookerjee Port) by improving the regime and navigability of the Bhagirathi-Hoogly river systems in 1975. The water-carrying capacity of Farakka canal is 40,000 cusecs.

18. Mangalam Irrigation Project is located in:

- (a) Andhra Pradesh
- (b) Kerala
- (c) Tamil Nadu
- (d) West Bengal

U.P. Lower Sub. (Pre) 1998

Ans. (b)

In Palakkad district of Kerala, 11 irrigation projects on Bharath Puzha river and its tributaries have been completed.

19. The Saran irrigation canal is drawn from the river

- (a) Son
- (b) Ganga
- (c) Kosi
- (d) Gandak

44th B.P.S.C. (Pre) 2000

Ans. (d)

A 740 metre long barrage has been built across the Gandak river in Balmiki Nagar, from where a few canals have been drawn. Saran Irrigation canal is one of them.

20. Indira Gandhi Canal has its source at :

- (a) Gandhi Sagar Dam
- (b) Bhakra Dam
- (c) Harike Barrage
- (d) Govind Ballabh Sagar Dam

U.P.P.C.S. (Pre) 1999

Ans. (c)

Indira Gandhi Canal originates at Harike Barrage near the confluence of Satluj and Beas rivers in Firozpur district of Punjab. The total length of the main canal system is 649 km. Sri Ganganagar, Bikaner, Jodhpur and Jaisalmer districts of western Rajasthan are mainly irrigated by Indira Gandhi Canal.

21. Harike Barrage (the source of water for Indira Gandhi Canal) is at the confluence of the rivers

- (a) Ravi and Beas
- (b) Jhelum and Chenab
- (c) Chenab and Satluj
- (d) Beas and Satluj

Uttarakhand U.D.A./L.D.A. (Mains) 2006

Ans. (d)

See the explanation of the above question.

22. Rajasthan (Indira) Canal originates from?

- (a) Ravi
- (b) Beas
- (c) Satluj
- (d) Chambal

U.P. P.C.S. (Pre) 1992

Ans. (b & c)

See the explanation of the above question.

23. The construction of the Indira Gandhi Canal started in the year 1958 and its origin is from :

- (a) Bhakra Dam on Sutlej river.
- (b) Rihand Valley Project on Rihand river.
- (c) Hirakund Project on Mahanadi.
- (d) Harike Dam on Satluj river.

R.A.S./R.T.S.(Pre) 2007

Ans. (d)

See the explanation of the above question.

24. Indira Gandhi Canal receives its water supply from

1. Beas
2. Chenab
3. Ravi
4. Satluj

Select the correct answer from the code given below :

Code :

- (a) 1 and 2
- (b) 2 and 3
- (c) 1, 2 and 3
- (d) 1, 3 and 4

U.P.P.S.C. (GIC) 2010

Ans. (d)

Indira Gandhi Canal receives its water supply from Satluj, Beas and Ravi.

25. Which one of the following river projects utilizes the water of the Pong barrage of Beas river?

- (a) Indira Gandhi Canal Project
- (b) Pochampad Project
- (c) Mayurakshi Project
- (d) Nagarjuna Sagar Project

U.P.U.D.A./L.D.A. (Pre) 2013

Ans. (a)

The Beas project is a joint project of Punjab, Haryana and Rajasthan states. Under this project, a dam was built on the Pong river to maintain water flow in the Indira Gandhi Canal during winter.

26. Which is the most well-developed and the oldest canal systems of the world in India?

- (a) Gang Canal
- (b) Sikri Canal
- (c) Indira Gandhi Nahar Pariyojna
- (d) Krishna-Godavari Canal System

R.A.S./R.T.S.(Pre) 2007

Ans. (a)

Gang Canal is one of the oldest irrigation systems in Rajasthan, completed in the year 1927. It was constructed by Maharaja Shri Gang Singh of Bikaner in 1927. The canal originates at Hussainiwala at Firozpur on the River Sutlej. The foundation stone of the Indira Gandhi Nahar Pariyojana was laid on 30 March, 1958 by the then Home Minister Govind Ballabh Pant. It has its source at Harike Dam in Punjab on the confluence of Satluj and Beas rivers.

27. Match List-I with List-II and choose the correct answer using the codes given below in the lists:

- | List-I
(Canals) | List-II
(Rivers) |
|--------------------------|---------------------|
| A. Upper Bari Doab Canal | 1. Sutlej |
| B. Sirhind Canal | 2. Yamuna |

C. Gurgaon Canal

3. Sutlej, Ravi, Beas

D. Indira Gandhi Canal

4. Ravi

Codes :

- | | A | B | C | D |
|-------|---|---|---|---|
| (a) 4 | 1 | 3 | 2 | |
| (b) 3 | 2 | 1 | 4 | |
| (c) 4 | 1 | 2 | 3 | |
| (d) 3 | 1 | 2 | 4 | |

UPPSC (Pre) 2024

Ans. (c)

The correct match is-

- | Canals | Rivers |
|-----------------------|--------------------|
| Upper Bari Doab Canal | Ravi |
| Sirhind Canal | Sutlej |
| Gurgaon Canal | Yamuna |
| Indira Gandhi Canal | Sutlej, Ravi, Beas |
- The Upper Bari Doab Canal has been taken out from the Ravi River in the state of Punjab. It is noteworthy that the region between the Beas and Ravi Rivers is called Bari Doab.
 - Sirhind Canal in the state of Punjab has been taken out from the Sutlej River.
 - The Gurgaon Canal has been taken out from the Yamuna River. This canal primarily provides irrigation facilities in Haryana.

The origin of the Indira Gandhi Canal is from the Harike Barrage, located downstream of the confluence of the Sutlej and Beas rivers in Punjab. From here, a 204 km. long feeder canal begins and at the end of this feeder canal, the 445 km. long Indira Gandhi main canal starts.

28. Gang canal, which is among the oldest canals, was constructed by Maharaja Ganga Singh Ji in –

- (a) 1930
- (b) 1927
- (c) 1944
- (d) 1932

R.A.S./R.T.S.(Pre) 2010

Ans. (b)

See the explanation of the above question.

29. Which of the following is/are the main objectives of Sharda Tributary Command Development Project ?

Select the correct answer using the code given below:

1. To increase agricultural production
2. To change the format of land use by multiple crop cultivation.
3. Reforms of land management.

Code :

- (a) Only 1 (b) Only 1 & 2
(c) Only 2 & 3 (d) All the above

U.P.P.C.S. (Pre) 2008

U.P.P.C.S. (Pre) 2002

U.P.U.D.A./L.D.A. (Pre) 2001

Ans. (d)

The Government of India initiated a centrally sponsored Command Area Development Programme (CADP) in December 1974 to improve irrigation potential utilization and optimize agricultural production from irrigated land through the integrated and coordinated approach of efficient water management. In tune with objectives of the programme, a number of components such as construction of field channels and field drains, enforcement of warabandi, land levelling and shaping, realignment of field boundaries/ consolidation of holdings, introduction of suitable cropping patterns, strengthening of extension services etc. were included in the programme and three projects were launched: (1) Sharda tributary (2) Ramganga & (3) Gandak. Thus, all three statements mentioned in the question are true. Hence, option (d) is the correct answer.

30. Which one of the following statements is not true?

- (a) Ghaggar's water is utilized in the Indira Gandhi Canal
(b) Narmada rises from Amarkantak region
(c) Nizam Sagar is situated on the Manjra river
(d) Penganga is a tributary of the Godavari

I.A.S. (Pre) 2000

Ans. (a)

The water of Ghaggar is not utilized in the Indira Gandhi Canal so statement (a) is wrong. Narmada river rises from Maikal range near Amarkantak in Anuppur district of Madhya Pradesh. So statement (b) is correct. The Nizam Sagar Dam is located in Kamareddy district of Telangana. Nizam Sagar is constructed across the Manjra river, a tributary of the Godavari and flows between the villages of Achampet and Banjapalle in Kamareddy district, Telangana. Thus statement (c) is also correct. The principal tributaries of Godavari river are Manjra, Pranhita, Indravati, Maner and Sabri river.

31. The Lower Ganga Canal is taken out of the Ganga at

- (a) Haridwar (b) Narora
(c) Bareilly (d) Kanpur

U.P.P.S.C. (GIC) 2010

Ans. (b)

The Lower Ganga Canal draws its water from the Ganga river at Narora (Bulandshahr). It was completed in 1878.

32. Which of the following canal systems irrigate areas of Bihar?

- (a) Upper Ganga Canal (b) Triveni Canal
(c) Sharda Canal (d) Eastern Yamuna Canal
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (b)

The Triveni Canal is made for irrigation in northwestern part of Bihar. This canal is related to irrigation project.

33. Hariyali is a new scheme for:

- (a) Second green revolution.
(b) Development of wasteland
(c) Forestation of desert land
(d) Development of grassland

Uttarakhand U.D.A./L.D.A. (Pre) 2003

Ans. (b)

The 'Hariyali' scheme was launched by the then Prime Minister Atal Bihari Vajpayee on 27 January, 2003 under watershed management project. This project is being run through 2.32 lakh panchayats of the country. This project encourages people to engage in afforestation.

34. 'Hariyali Yojana' is related to

- (a) Soil Management (b) Air Management
(c) Crop management (d) Water Management

U.P.P.C.S. (Mains) 2013

Ans. (d)

See the explanation of the above question.

35. What are the benefits of implementing the 'Integrated Watershed Development Programme?

1. Prevention of soil runoff
2. Linking the country's perennial rivers with seasonal rivers
3. Rainwater harvesting and recharge of groundwater levels
4. Regeneration of natural vegetation

Select the correct answer using the code given below.

- (a) 1 and 2 only (b) 2, 3 and 4 only
(c) 1, 3 and 4 only (d) 1, 2, 3 and 4

I.A.S. (Pre) 2014

Ans. (c)

Till 1, April 2008, the Ministry of Rural Development, Government of India implemented three watershed programmes, namely the Integrated Wastelands Development Programme the Drought Prone Areas Programme and Desert Development Programme. Since 1st April, 2008 they have been brought under a comprehensive programme named Integrated Watershed Management Programme (IWMP), to be implemented under Common Guidelines on Watershed Development, 2008. The main objectives of the IWMP are to restore the ecological balance by harnessing, conserving and developing degraded natural resources such as soil, vegetative cover and water. The outcomes are the prevention of soil run-off, regeneration of natural vegetation, rainwater harvesting and recharging of the groundwater level. This enables multi-cropping and the introduction of diverse agro-based activities, which help to provide sustainable livelihoods to the people residing in the watershed area. Thus, statement 2 is not included in the benefits of implementing the 'Integrated Watershed Development Programme'.

36. Consider the following pairs :

- | Programme/Project | Ministry |
|---|---------------------------------------|
| 1. Drought-Prone Area Programme | : Ministry of Agriculture |
| 2. Desert Development | : Ministry of Environment and Forests |
| 3. National Watershed Development Project for Rainfed Areas | : Ministry of Rural Development |

Which of the above pairs is/are correctly matched?

- (a) 1 and 2 only (b) 3 only
(c) 1, 2 and 3 (d) None

I.A.S. (Pre) 2014

Ans. (d)

The Department of Land Resources implemented 3 watershed programmes, viz. Integrated Wastelands Development Programme (IWDP), Drought Prone Areas Programme (DPAP) and Desert Development Programme (DDP). These were merged under a comprehensive programme named Integrated Watershed Management Programme (IWMP) in February, 2008. The Department of Land Resources is a part of the Ministry of Rural Development, Government of India. National Watershed Development Project for Rainfed Areas (NWDPA) is an initiative programme of the Ministry of Agriculture (MoA).

37. Which of the following is/are the advantage/advantages of practising drip irrigation?

1. Reduction in weed
2. Reduction in soil salinity
3. Reduction in soil erosion

Select the correct answer using the code given below :

- (a) 1 and 2 only
(b) 3 only

(c) 1 and 3 only

(d) None of the above is an advantage of practising drip irrigation

I.A.S. (Pre) 2016

Ans. (c)

Drip irrigation is a form of irrigation that saves water and fertilizer by allowing water to drip slowly to the roots of many different plants. Advantages of practising drip irrigation are: maximum use of available water, maximum crop yield, reduction in weed, reduction in soil erosion, low labour cost, no runoff of fertilizers, less evaporation of water, improved seed germination etc. Thus, statements (1) and (3) are correct and (c) is the correct answer.

38. Buckingham canal is situated :

- (a) On Kalinga Coast (b) On Malabar Coast
(c) On Coromandel Coast (d) On Kutch Coast

Chhattisgarh P.C.S. (Pre) 2019

Ans. (c)

Buckingham canal is a freshwater navigation canal with a length of 796 km, paralleling the Coromandel coast from Kakinada district of Andhra Pradesh to Parangipettai near Chidambaram in Tamil Nadu. It connects most of the natural backwaters along the coast to Chennai port.

39. Which one of the following sector has greater consumption share of water in India?

- (a) Agricultural Sector (b) Domestic Sector
(c) Mining Sector (d) Industrial Sector

Jharkhand P.C.S. (Pre) 2021

Ans. (a)

In India, the maximum use of water through irrigation is in the agricultural sector. It is to be noted that the total water availability obtained through rainfall in India is about 3880 billion cubic meters (bcm) per year.

40. Chambal Canal Irrigation Project is related with :

- (a) Sindh Basin (b) Ganga Basin
(c) Yamuna Basin (d) Tapti Basin

M.P.P.C.S. (Pre) 2020

Ans. (b & c)

Chambal Canal Irrigation Project is related to Yamuna basin. Chambal is a tributary of the Yamuna, but it is also a sub tributary of the Ganges and is a part of the Ganges basin as a whole. Therefore, according to the larger basin, it is related to the Ganges basin, while as a smaller basin it is related to the Yamuna basin. If the most appropriate answer is to be selected, then Yamuna basin is the most appropriate answer, but here both options (b) and (c) seem to be correct on the basis of the vagueness of the question.